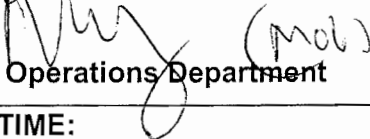


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Emergency Diesel Generator		
TASK:	Unload and stop an EDG		
TASK NUMBER:	N0645060104		
JPM NUMBER:	13-01 NRC IP-i		
ALTERNATE PATH:	☐	K/A NUMBER:	064 A2.03
APPLICABILITY:		IMPORTANCE FACTOR:	3.1 3.1
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:		In Plant / Simulate	
REFERENCES: S2.OP-SO.DG-0003, Rev. 41 (checked 10-15-14)			
TOOLS AND EQUIPMENT: None			
VALIDATED JPM COMPLETION TIME:		10 min	
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:		N/A	
Developed By:	G Gauding Instructor	Date:	9-24-14
Validated By:	S Bickhart SME or Instructor	Date:	10-15-14
Approved By:	 Training Department	Date:	10-23-14
Approved By:	 Operations Department	Date:	10-23-14
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

**TASK
NUMBER:** N0645060104

**INITIAL
CONDITIONS:**

2C EDG is running loaded in parallel with 2C Vital bus for maintenance. It is loaded to 2550 KW, 1500 KVAR OUT.

INITIATING CUE:

Unload and shut down the 2C EDG IAW S2.OP-SO.DG-0003, 2C Diesel Generator Operation, Sections 5.8 and 5.9

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Lower DG real load incrementally to less than 200 KW.
2. Lower DG reactive load incrementally to 50-100 KVAR.
3. Open DG output breaker.
4. Stop DG and return Gen Loading SW to AUTO.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS
					(Required for UNSAT Evaluation)
		Provide candidate with marked up procedure.	Give operator marked up copy of S2.OP-SO.DG-0003, 2C Diesel Generator Operation, Rev. 41 .		
	NOTE	The GENERATOR CIRCUIT BREAKER SWITCH should be placed in the TRIP position as soon as EDG Load is reduced to 200 KW.	Ensure that operator simulates tripping the EDG breaker when cue is given later in JPM that load is <200KW.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* *	5.8.1	IF 2C EDG is paralleled on 2C 4KV Vital Bus, <u>THEN</u> concurrently PERFORM the following: • ADJUST SPEED CONTROL SWITCH (GS) to gradually lower load to 200 KW. • ADJUST VOLTAGE CONTROL SWITCH (VCS) to gradually lower KVAR to between 50-100 KVAR OUT.	Operator locates SPEED CONTROL SWITCH (GS) and VOLTAGE CONTROL SWITCH, and simulates alternately goes to LOWER on the switches while monitoring Generator KW and KVAR. <u>CUE</u>: Each time operator simulates going to lower on SPEED CONTROL SWITCH (GS), give operator reading which is 300KW less than previous. <u>CUE</u>: Each time operator simulates going to lower on VOLTAGE CONTROL SWITCH (VCS), give operator reading which is 200KVAR less than previous. <u>NOTE</u>: IF operator simulates going to RAISE on either switch, then CUE a reading that is HIGHER in the same amount as proper operation above.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.8.3	OPEN 2C DIESEL GENERATOR BREAKER by performing one of the following: • OPEN 2CD1AX6D, 2C DIESEL GENERATOR BREAKER. (64' Swgr Rm, or • PRESS 2C BREAKER OPEN pushbutton (2CC3), or • PLACE 2C-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS), to TRIP (2C D/G Control Panel).	Simulates opening 2C DIESEL GENERATOR BREAKER by placing 2C-DF-GCP-3 (BCS) to TRIP position. CUE: Green indicating light above switch illuminates, red light extinguishes. Generator KW and KVAR indicate 0. Generator Voltage indicates 4160 VAC.		
	5.8.4	ALLOW Diesel to run unloaded for ≥ 3 minutes prior to stopping the EDG.	CUE: 3 minutes has elapsed.		
	5.8.5	<u>IF</u> diesel unloading was directed by Step 5.9.3F <u>OR</u> Step 5.10.4E, <u>THEN RETURN</u> to Step 5.9.4 <u>OR</u> Step 5.10.5.	Marks step N/A		
	5.9.1	<u>IF</u> Diesel Generator was operated without electrical load for <15 minutes, <u>THEN ALLOW</u> Diesel to run unloaded until total runtime is ≥ 15 minutes <u>OR</u> for a period of time specified by a pre-planned Maintenance activity, prior to stopping Diesel.	Marks step N/A		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.9.2	<u>IF</u> Diesel Generator was operated without electrical load for ≥ 2 hours, <u>AND</u> the diesel is capable of loading, <u>THEN:</u> [C0300] A. Slowly LOAD Diesel, over 15 minutes, to 1500 KW at 1125 KVAR for minimum of 1 hour IAW Section 5.4 of this procedure. B. UNLOAD Diesel IAW Section 5.8 of this procedure.	Marks step N/A		
	5.9.3	<u>IF</u> Diesel Generator was operated without electrical load for ≥ 2 hours....	Marks step N/A		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.9.4	<u>IF</u> 2C-DF-GCP-1 2C DIESEL GEN LOADING SW is in MANUAL (DROOP), <u>THEN</u>: A. ADJUST Diesel Generator speed as necessary to achieve 900 rpm using SPEED CONTROL SWITCH (GS). B. CHECK frequency indicated on 2CC3 bezel is 58.94-60.36 Hz. C. ADJUST Generator terminal voltage as necessary to achieve 4160 volts using VOLTAGE CONTROL SWITCH (VCS) D. CHECK voltage indicated on 2CC3 is 4.023-4.306 Kv.	<u>CUE</u> IF ASKED: 2C DIESEL GEN LOADING SWITCH is in MANUAL) <u>CUE</u>: DG speed indicates 905 rpm. Operator simulates placing SPEED CONTROL SWITCH (GS). to lower, then <u>CUE</u>: DG speed indicates 900 rpm. Calls control room to check frequency. <u>CUE</u>: frequency is 60.03 HZ. <u>CUE</u>: Generator voltage is 4140. Simulates raising VOLTAGE CONTROL SWITCH (VCS). <u>CUE</u>: Generator voltage is 4160. Operator calls control room to check voltage on 2CC3 bezel. <u>CUE</u>: Voltage is 4200.		
*	5.9.5	PLACE 2C-DF-SS, 2C DG STOP/START SWITCH in STOP.	Simulates placing 2C-DF-SS 2C DG STOP/START SWITCH to STOP, and verifies speed lowering. <u>CUE</u>: Diesel is stopping.		
	5.9.6	<u>IF</u> the Diesel Generator does not shutdown...	Marks NA.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Diesel Generator

TASK: Unload and stop an EDG

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.9.7	<u>IF</u> 2C-DF-GCP-1, 2C DIESEL GEN LOADING SW is in MANUAL (DROOP), <u>THEN PLACE</u> 2C-DF-GCP-1 2C DIESEL GEN LOADING SW in AUTO (ISOCR).	Simulates placing 2C-DF-GCP-1 2C DIESEL GEN LOADING SW in AUTO. CUE: AUTO light illuminates, MANUAL light extinguishes, GENERATOR LOADING IN DROOP MODE (B-9) alarm clears.		

TERMINATION: After placing switch in AUTO, state "JPM is complete."

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

1. Task description and number, JPM description and number are identified.
2. Knowledge and Abilities (K/A) references are included.
3. Performance location specified. (in-plant, control room, or simulator)
4. Initial setup conditions are identified.
5. Initiating and terminating Cues are properly identified.
6. Task standards identified and verified by SME review.
7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 41 Date 10/15/14
9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor:  BICKHART

Date: 10/15/14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

**INITIAL
CONDITIONS:**

2C EDG is running loaded in parallel with 2C Vital bus for maintenance. It is loaded to 2550 KW, 1500 KVAR OUT.

INITIATING CUE: Unload and shut down the 2C EDG IAW S2.OP-SO.DG-0003, 2C Diesel Generator Operation, Sections 5.8 and 5.9